

Revision

Based on

Unit 1: magnetic effect of electric current

Unit 2: electromagnetic induction

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Revision sheet (Unit 1)

A) What is meant by?

1) Magnetic flux density

The magnetic force acting on a conductor of length 1 meter carrying an electric current of intensity 1 ampere placed perpendicular to magnetic field

OR (The number of magnetic flux lines passing normally through unit cross section area)

2) Tesla

The magnetic flux density when a force of 1 Newton acting on a conductor of length 1 meter carrying an electric current of intensity 1 ampere placed perpendicular to magnetic field

OR (The magnetic flux density when the number of magnetic flux lines passing normally through unit cross section area is 1 Weber)

3) Neutral point

The point at which the total magnetic flux density vanishes

4) Magnetic dipole moment

It is a vector emanating from North Pole of the coil perpendicular to its area

5) Sensitivity of the galvanometer

The scale deflection per unit current intensity passing through its coil

6) Shunt (R_s)

A small resistance connected **in parallel** with coil's galvanometer to convert it into ammeter to measure high current intensity

7) Multiplier (R_m)

A large resistance connected **in series** with coil galvanometer to convert it into voltmeter to measure high potential difference

8) Magnetic flux density = 0.3 T.

The magnetic Force acting on a conductor of length 1 meter carrying an electric current of intensity 1 ampere placed perpendicular to magnetic field is **0.3 N**.

OR (The number of magnetic flux lines passing normally through unit cross section area is 0.3weber).

B) Give reasons for each of the following:

1) It is advisable to live away from high voltage towers.

Due to the relation $\frac{\mu I}{2\pi d}$

As the magnetic field increases by increasing the current (causes bad effect on human health) and to decrease this effect we can live away from high voltage towers. Where the magnetic field decreases by increasing the distance of point from the wire ($B \propto 1/d$)

2) The magnetic flux density along the axis of a solenoid increases if we insert inside it a bar of soft iron.

Because the soft iron has high magnetic permeability and concentrates the magnetic lines inside the solenoid, thus the magnetic flux density increases.

3) In a solenoid when the separation between each two turns decreases, the magnetic flux density at a point inside the coil on its axis will increase

Because the length of the axis of the solenoid decreases and there is an inverse relation between the magnetic flux density and length of the solenoid ($B \propto 1/l$)

4) A straight wire carrying a current placed in a magnetic field, the wire did not move.

According to the following relation

$$F = B I \sin \theta$$

The straight wire is placed in parallel with the magnetic field, so the angle between the wire and magnetic field is zero

C) Mention the necessary conditions to obtain:

The neutral point at midpoint between two parallel straight wires of each carrying current	1) The two current are in the same direction 2) The two currents are equal ($I_1 = I_2$)
The force acting on straight wire carrying a current placed in a uniform magnetic field equal zero	The wire is placed parallel to magnetic field The angle between the magnetic flux and the wire is 0° ($\theta = 0$)
The force acting on straight wire carrying a current placed in a uniform magnetic field equal maximum	The wire is placed perpendicular to magnetic field The angle between the magnetic flux and the wire is 90° ($\theta = 90^\circ$)
The force between two parallel wires each of them carrying current is attractive	The two currents are in same direction
The force between two parallel wires each of them carrying current is repulsive	The two currents are in opposite direction
The torque acting on rectangular coil carrying a current placed in a magnetic field equal maximum value	The coil's plane parallel to magnetic field The angle between the magnetic flux and normal to coil's plane is 90° ($\theta = 90^\circ$)
The torque acting on rectangular coil carrying a current placed in a uniform magnetic field equal zero	The coil's plane normal to magnetic field The angle between the magnetic flux and normal to coil's plane is 0° ($\theta = 0^\circ$)

D) Mention one function or application for each of the following

Ampere's right hand rule	To determine the direction of magnetic field resulting from an electric current in a wire or a solenoid
Right screw hand rule	To determine the direction of magnetic field resulting from an electric current in a circular coil
Fleming's left hand rule	To determine the direction of magnetic force acting on a wire carrying current placed in a magnetic field

E) Mention the scientific idea for each of the following

1) Moving coil galvanometer 2) dc ammeter	The magnetic torque acting on a coil carrying current placed in a uniform magnetic field. OR magnetic effect of electric current
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F)Mention the factors affecting each of the following

Magnetic flux density due to current passing through straight wire: (B)	1) electric current intensity ($B \propto I$) 2) distance between wire and point ($B \propto 1/d$)
Magnetic flux density due to current passing through circular coil: (B)	1) electric current intensity ($B \propto I$) 2) number of turns ($B \propto N$) 3) radius of coil ($B \propto 1/r$)
Magnetic flux density due to current passing through solenoid(B)	1) electric current intensity ($B \propto I$) 2) number of turns ($B \propto N$) 3) length of solenoid ($B \propto 1/L$)
Force acting on a wire carrying current placed in a magnetic field:(F)	1) magnetic flux density ($F \propto B$) 2) electric current intensity ($F \propto I$) 3) length of wire ($F \propto L$) 4) angle between wire and field ($F \propto \sin \theta$)
The mutual force between two parallel wires of each carrying current	1) the current intensity in 1 st wire ($F \propto I_1$) 2) the current intensity in 2 nd wire ($F \propto I$) 3) the distance between the two parallel wires ($F \propto 1/d$)
Magnetic torque acting on a coil carrying current placed in a magnetic field: (τ)	1) magnetic flux density ($\tau \propto B$) 2) electric current intensity ($\tau \propto I$) 3) area of coil ($\tau \propto A$) 4) number of turns ($\tau \propto N$) 5) angle between perpendicular to coil and field ($\tau \propto \sin \theta$)

F)State the properties of each of the following:

Magnetic field at a point from a wire carrying current	1) The magnetic lines of force around a straight wire carrying current are concentric circles whose centers lie on the wire, If the current passing through the wire increases , the concentric circles more crowded 2) The circular magnetic flux lines are closer together near the wire and farther apart from each other as the distance from the wire increases
Magnetic field at center of circular coil carrying current	The magnetic field produced is very similar to that of a short bar magnet 1) The flux lines are circular surrounding the two sides 2) The flux lines are no longer circular near the center 3) The flux lines become straight and parallel lines perpendicular to the plane of the coil at the center of the coil. This means that the magnetic field in this region is uniform.
Magnetic field at a point along the solenoid carrying current	The magnetic field produced by a current carrying solenoid is similar to the magnetic field produced by a bar magnet 1) The lines of magnetic flux outside the solenoid are closed loops moving from the North Pole to the South Pole 2) The lines of magnetic flux through the middle of the solenoid (In a solenoid) are straight and parallel to the axis, so the magnetic field is uniform

Neutral point	1)The total magnetic flux density at this point vanishes (The two magnetic field produced when current passing through the two conductors are equal in magnitude and opposite in direction ($B_1 = B_2$))
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G) Compare between each pair:

Straight wire	Circular coil	Solenoid
$B = \frac{\mu I}{2\pi d}$	$B = \frac{\mu NI}{2r}$	$B = \frac{\mu NI}{L}$
The magnetic field lines take concentric circles around the wire which are closer together near the wire.	Similar to that of a short bar magnet. The magnetic flux lines at the center of the loop are straight parallel lines and perpendicular to the coil.	similar to that as a large bar magnet Inside the solenoid the magnetic flux lines are parallel to the axis of the solenoid
Ampere right hand rule	Right hand screw rule	Ampere right hand rule
Ampere's right hand rule	screw right hand rule	Fleming left hand rule
Used to determine the direction of magnetic field resulting from current passing through a wire or solenoid	Used to determine the direction of magnetic field resulting from current passing through a circular coil or solenoid	Used to determine the direction of magnetic force acting a wire carrying current placed in a uniform magnetic field or solenoid

Prove that the magnetic force (F) acting on a straight wire of length (l) placed in a uniform magnetic field of flux density (B) is given by the following relation

$$F = B I l \sin\theta$$

The proof

$$F \propto B$$

$$F \propto I$$

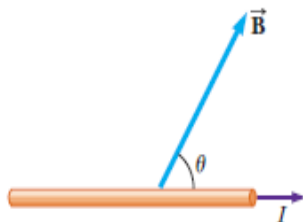
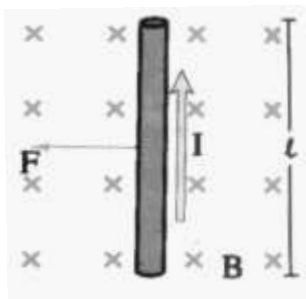
$$F \propto l$$

$$F = \text{constant} \times B I l$$

$$\text{Constant} = \sin\theta$$

$$F = B I l \sin\theta$$

Where θ is the angle between the wire and magnetic field



Prove that the mutual force between two parallel wires each carrying current is given by

$$F = F_1 = F_2 = \frac{\mu I_1 I_2 L}{2\pi d}$$

The proof

$$F_1 = B_2 I_1 L$$

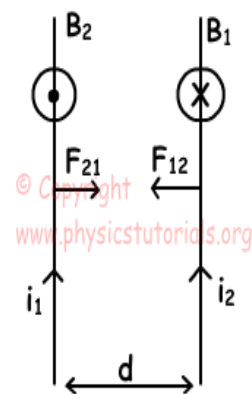
$$F_2 = B_1 I_2 L$$

$$B = \frac{\mu I}{2\pi d}$$

$$F_1 = \left(\frac{\mu I_2}{2\pi d}\right) I_1 L = \frac{\mu I_1 I_2 L}{2\pi d}$$

$$F_2 = \left(\frac{\mu I_1}{2\pi d}\right) I_2 L = \frac{\mu I_1 I_2 L}{2\pi d}$$

$$F = F_1 = F_2$$



Mention only the name of one apparatus whose function is based on each of the following and mention one use of it:

The scientific base	The apparatus
1) the torque acting on a moving coil placed in a magnetic field resulting from the following of current through it	1) the galvanometer
	2) DC motor

When does each of the following values become equal to maximum?

1) the force acting on straight wire carrying a current placed in a magnetic field when the wire is placed perpendicular to the magnetic field

θ (the angle between the wire and magnetic field) = 90°

2) The magnetic torque acting on rectangular coil carrying a current placed in a magnetic field

when the coil is placed parallel to the magnetic field

θ (the angle between the normal to coil's plane and magnetic field) = 90°

When does each of the following values become equal to zero?

1) The magnetic force acting on straight wire carrying a current placed in a magnetic field

when the wire is placed parallel to the magnetic field

θ (the angle between the wire and magnetic field) = zero

2) The magnetic torque acting on rectangular coil carrying a current placed in a magnetic field

when the coil is placed perpendicular to the magnetic field

θ (the angle between the normal to coil's plane and magnetic field) = zero

How to increase each of the following:

1) Magnetic flux density due to straight wire

a) by increasing the electric current intensity passing through the wire

b) by decreasing the distance between the wire and the point

2) Magnetic flux density due to circular coil

a) by increasing number of turns of the coil

b) by increasing the electric current intensity passing through the coil

c) by decreasing the radius of coil

3) Magnetic flux density due to solenoid

a) by increasing number of turns of the solenoid

b) by increasing the electric current intensity passing through the coil

c) by decreasing the length of solenoid

4) Force acting on a wire in a magnetic field

a) by increasing the magnetic flux density

b) by increasing the electric current intensity passing through the wire

c) by increasing the length of wire

5) Mutual force between two parallel wires

- a) by increasing the current passing through each wire
- b) by decreasing the distance between the two wires

6) Magnetic torque acting on a rectangular coil carrying current in a magnetic field

- a) by increasing the magnetic flux density
- b) by increasing the electric current intensity passing through the coil
- c) by increasing the area of coil
- d) by increasing number of turns of the coil

What happens.....? :(Explain each of them briefly)

OR What are the results produced in each of the following

when The electric current passing through the straight wire increases (with respect to the magnetic field density at a point from the wire)

Ans: the magnetic field at a point from the wire will increase

The reason : according to the relation $B = \mu \frac{I}{2\pi d}$

Greater the current in the wire, stronger will be the magnetic field produced. Where the magnetic field density is directly proportional to the current intensity

when the radius (diameter) of the circular coil increases to its double value (with respect to the magnetic field density at the center)

Ans: the magnetic field at a point from the wire will decrease to its half value

The reason : according to the relation $B = \mu \times \frac{NI}{2r}$

the magnetic field density is inversely proportional to the radius (diameter) of the circular coil

When a soft iron bar inserted in a solenoid carrying current(with respect to the magnetic field density at a point along the axis)

Ans: the magnetic field at a point the axis of the solenoid will increase

The reason : according to the relation $B = \mu \frac{NI}{L}$

Because the soft iron has high permeability, so the magnetic flux lines will be concentrated. the magnetic field density is directly proportional to magnetic permeability

Placing two wires parallel and carrying electricity in the same direction.

Ans: Attraction between the two wires and the neutral point formed between them

Placing two wires parallel and carrying electricity in opposite directions.

Ans: Repulsion between the two wires the neutral point formed outside the two wires

A solenoid of length (L) and number of turns (N) is connected to a battery of emf (V_B) and negligible internal resistance. What would happen to the magnetic flux density at a point inside the coil on its axis in the following cases (mention the reason in each case?)

1) When the separation between each two turns is reduced to its half

1) Ans : the magnetic flux density will be doubled

The reason: because the length of the solenoid decreases to its half value, and the magnetic flux density is inversely proportional to the length of the solenoid

$$B = \frac{\mu NI}{L}$$

Problems: (magnetic effect of electric current)

The distance between two parallel straight wires is 10 cm; one of them carries a current of 2A while the other carries a current of 3A in the same direction. Find the position of the neutral point. If one of the two currents in the previous problem is reversed and a third wire of length 10 cm carrying a current of the 5A, is placed at the previous neutral point. Find the force acting on it.	At the natural point $B_1 = B_2$ $\frac{\mu I_1}{2\pi d} = \frac{\mu I_2}{2\pi (10 - d)}$ $\frac{2}{d} = \frac{3}{10 - d}$ $20 - 2d = 3d$ $d = 4\text{cm}$ the nutral point at 4cm from the wire carries 2A When one of the current is reversed $B_t = B_1 + B_2$ $B_t = \frac{\mu I_1}{2\pi d_1} + \frac{\mu I_2}{2\pi d_2}$ $B_t = 2 \times 10^{-7} \left(\frac{2}{0.04} + \frac{3}{0.06} \right) = 2 \times 10^{-5} \text{T}$ $F = B I l$ $F = (2 \times 10^{-5})(5)(0.1) = 10^{-5} \text{ N}$
A circular coil of a diameter 10cm ,(N) turn ,carrying a current of intensity (I) and producing a magnetic field in its center , if the coil becomes a solenoid .calculate the length of the solenoid when the magnetic flux density on the solenoid axis equals (1/ 4) the flux density in the center of circular coil	$\frac{B_{\text{coil}}}{B_{\text{solenoid}}} = \frac{l}{2r}$ $B_{\text{solenoid}} = (1/4) B_{\text{coil}}$ $4B_{\text{solenoid}} = B_{\text{coil}}$ $\frac{4 B_{\text{solenoid}}}{B_{\text{solenoid}}} = \frac{l}{10}$ $l = 40 \text{ cm}$
A metallic wire is wrapped in the form of circular coil of radius 7cm and 4 turns carries an electric current, which generates a magnetic field at the center of flux density equal $3.52 \times 10^{-5} \text{ Wb/m}^2$. Calculate the current through the coil.	$B_{\text{coil}} = \frac{\mu I N}{2r}$ $3.52 \times 10^{-5} = \frac{43.52 \times 10^{-5} \times 10^{-7} \times I \times 4}{2 \times 0.07}$ $I = 0.98 \text{ A}$
A rectangular coil of dimensions 20cm x 10cm and of 200 turns placed in a regular field of magnetic flux density of 0.4 T. current of 3A is passing through the coil. Calculate the torque acting on the coil in the following two cases: a) When the coil plane subtends an angle 60° to the direction of the field; b) When the coil plane is perpendicular to the direction of the field.	$\tau = B I A N \sin\theta$ $\tau = 0.4 \times 3 \times 20 \times 10 \times 10^{-4} \times 200 \times \sin 30$ $\tau = 2.4 \text{ N.m}$ $\tau = B I A N \sin\theta$ $\tau = 0.4 \times 3 \times 20 \times 10 \times 10^{-4} \times 200 \times \sin 0$ $\tau = 0 \text{ N.m}$ θ:the angle between normal to coil's plane and magnetic field
A circular coil of 200 turns, radius 10 cm carries a current of 5 A. It is suspended in a uniform magnetic field of 0.4 T. the field lines making and angle 60° with the plane of the coil. Calculate the magnitude of the torque that must be applied on it then the value of magnetic dipole moment	$\tau = B I A N \sin\theta$ $\tau = 0.4 \times 5 \times \pi \times 10^2 \times 10^{-4} \times 200 \times \sin 30$ $\tau = 6.28 \text{ N.m}$ $m_d = I A N$ $m_d = 5 \times \pi \times 10^2 \times 10^{-4} \times 200$ $m_d = 31.4 \text{ A.m}^2$

Revision sheet (unit 2)

A) What is meant by?

1) electromagnetic induction

An induced emf and also an induced current are generated in the conductor by a changing magnetic field (**magnetic flux**) linking with the conductor

2) Faraday's law

The magnitude of induced emf is directly proportional to rate of change of magnetic flux

3) Lenz's law

The induced current must be in a direction such as to oppose the change producing it.

4) Mutual Induction between two coils phenomenon

It's the electromagnetic effect takes place between two coils due current variation in one of them (primary coil) and an induced emf generated in the other coil (secondary coil).

5) Self induction in a coil phenomenon

Induced emf generated in a coil due to the current in the same coil is changed, the magnetic flux link also changes.

6) Eddy current

It is induced current that circulates in closed paths due to the change in magnetic flux through a solid conductor.

7) The mutual inductance between two coils = 0.5 Henry

Induced emf generated in the secondary coil when the current intensity passing in the primary coil changes by a rate of 1 A/sec is **0.5 volts**

8) The self inductance of a coil = 0.2 Henry

Induced emf generated in the coil when the current intensity passing in it changes by a rate of 1 A/sec. is **0.2 volt**

9) The Henry

It numerically equals an emf of 1 V induced in the coil by self induction when the rate of change of current intensity in the same coil is 1 Ampere/second.

10) The induced emf produced in a coil when the current passing through it changes by a rate of 1 A/sec equal to 0.5 volts

The self inductance of this coil = **0.5 Henry**

11) The effective value of an alternating current = 3.5 A

The value of the intensity of direct current which produces the same rate of heat energy in a resistor as that produced by the alternating current is **3.5A**.

12) Efficiency of an electric transformer = 90%

The ratio between energy gained from the secondary coil to that given to the primary coil in same time is **0.9**.

OR

The ratio between the power taken from secondary coil to the power given to the primary coil of transformer is **0.9**

B) Give reasons for each of the following:

1) When a magnet moving towards or away from a coil ,an induced emf is produced in the coil

Because when the is moving towards or away from a coil there is a change in magnetic flux linking with the coil

2) The induced emf generated in the coil increases when an iron core is used

Because the iron has high magnetic permeability so it will concentrate the magnetic flux lines (the magnetic flux will increases) and according to the relation $\text{emf}_{\text{induced}} \propto \frac{\Delta \phi}{\Delta t}$ so the induced emf will increases

3) Self-inductance of a coil increases when an iron core is used.

As it increases the permeability of the solenoid and concentrates the magnetic lines inside it, thus the magnetic flux density increases.

4) The eddy currents are useful in melting metals.

Eddy currents produce heat as it is used to form induction furnaces to melt metals.

5) A straight wire may move between magnetic poles, and no induced emf produced.

Because the straight wire is moving in parallel with magnetic field, so the angle between the direction of motion and direction of magnetic field is zero ($\theta = 0$, $\sin 0 = 0$)

$$\text{emf}_{\text{ind}} = B l v \sin \theta.$$

6) The induced emf in a dynamo coil has maximum value when the coil plane is parallel to the field.

According to the relation $\text{emf} = A B N 2\pi f \sin \theta$

When coil is parallel to the field, the angle between the perpendicular to the coil and the field is 90° which means maximum induced emf. ($\sin 90 = 1$)

C) Mention the necessary conditions to obtain:

The instantaneous induced emf generated in rectangular coil (dynamo) rotates in a uniform magnetic field equal zero	The coil's plane perpendicular to magnetic field OR The angle between the magnetic flux and normal to coil's plane is 0° ($\theta = 0^\circ$)
The instantaneous induced emf generated in rectangular coil rotates in a magnetic field equal maximum	The coil's plane parallel to magnetic field OR The angle between the magnetic flux and normal to coil's plane is 90° ($\theta = 90^\circ$)
The average induced emf generated in a coil equal zero	If the coil rotates one complete cycle from beginning of rotation (zero position)
induced emf generated in a straight wire rotates in a uniform magnetic field equal zero	If the wire is moved parallel to the direction of magnetic field OR The angle between the direction of magnetic field and direction of motion is 0° ($\theta = 0^\circ$)
induced emf generated in a straight wire rotates in a uniform magnetic field equal maximum	If the wire is moved perpendicular to the direction of magnetic field OR The angle between the direction of magnetic field and direction of motion is 90° ($\theta = 90^\circ$)
Step up transformer	Number of turns of the secondary coil must be greater than that of primary coil
Step down transformer	Number of turns of the secondary coil must be less than that of primary coil

D)Mention one function or application for each of the following

Lenz's Law	It is used to determine the direction of induced emf generated in the coil.
Self Induction	It is used to produce large emf which is used in Fluorescent lamp
Eddy current	It is used in the induction furnaces to melt metals.
Fleming's right hand rule	To determine the direction of induced current in a straight wire placed in a uniform magnetic field
AC generator	Convert the mechanical energy into electrical energy
The two carbon brushes in the dynamo	To make the induced current in the coil passes to the external circuit
Transformer	1) It's used to transfer the electric energy from the generators at electric power stations 2) It's used to step up or to step down an AC voltage.

E)Mention the scientific idea for each of the following

AC dynamo	Electromagnetic induction
DC motor	The magnetic torque acting on a coil carrying current placed in a magnetic field
Electric transformer	Mutual induction or electromagnetic induction
Induction furnaces	Eddy current

F)State the properties of each of the following:

The output current from AC generator	1) It changes its direction each half cycle and its intensity periodically 2) It flows in two opposite directions in a circuit 3) it can be converted into a DC
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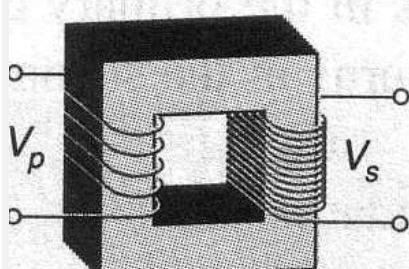
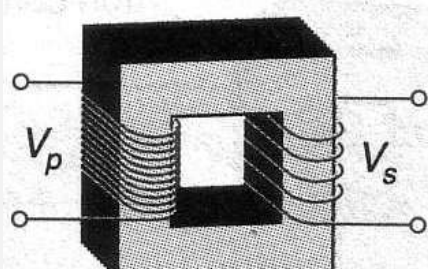
G)Mention the factors affecting each of the following

The average induced emf generated in a coil	1) The number of turns ($\text{emf} \propto N$) 2) The rate of change of magnetic flux ($\text{emf} \propto \frac{\Delta\phi}{\Delta t}$)
Mutual induction coefficient (M)	1) Presence of soft iron core inside the coil 2) Number of turns of the coil and the volume of the two coil (Geometry of the two coils) 3) Distance between the two coils
Self induction coefficient of a coil (L)	1) Presence of soft iron core inside the coil 2) Number of turns of the coil and the volume of the coil (Geometry of the coil) 3) Distance between the turns of the coil
The induced emf. in a straight wire moving in a magnetic field (emf)	1) Magnetic flux density ($\text{emf} \propto B$) 2) Length of wire ($\text{emf} \propto l$) 3) Velocity of wire ($\text{emf} \propto v$) 4) Angle between wire and field ($\text{emf} \propto \sin \theta$)
Maximum induced emf in the dynamo (emf_{max})	1) Number of turns of coil ($\text{emf} \propto N$) 2) Magnetic flux density ($\text{emf} \propto B$) 3) Area of coil ($\text{emf} \propto A$) 4) frequency ($\text{emf} \propto f$)

H) Compare between each pair:

A back induced emf	A forward induced emf
Switching on the current in primary coil	Switching off the current in primary coil
Increasing the current intensity in primary coil by using rheostat.	Decreasing the current intensity in the primary coil by using rheostat
Putting the primary coil near to or inside secondary coil	Withdrawing primary coil away from secondary coil
Its direction opposes the direction of the original current	Its direction same as the direction of the original current

Fleming's right hand rule	Lenz's rule
Used to determine the direction of induced current in a straight wire moving in a uniform magnetic field	It is used to determine the direction of induced emf generated in the coil.

Step up transformer	Step down transformer
1) Step up the AC voltage. 2) Decrease the current intensity	1) Step down the AC voltage. 2) Increase the current intensity
$V_P < V_S$	$V_P > V_S$
$N_P < N_S$	$N_P > N_S$
$I_P > I_S$	$I_P < I_S$
	

Mention only the name of one apparatus whose function is based on each of the following and mention one use of it:

The scientific base	The apparatus	The use
2) Electromagnetic induction	AC generator (dynamo)	Convert the mechanical energy into electrical energy
	Electric transformer	Step up or step down Ac voltage
3) Mutual induction between two coils	Electric transformer	Step up or step down Ac voltage
4) eddy current	Induction furnaces	Melting metals

When does each of the following values become equal to maximum?

1) Induced emf generated in a conductor (straight wire) moves in a uniform magnetic field

when the conductor moves perpendicular to the magnetic field lines

θ (the angle between the direction of wire and magnetic field) = 90°

2) induced emf generated in rectangular coil (generator) rotates in a uniform magnetic field

when the coil is placed parallel to the magnetic field

θ (the angle between the normal to coil's plane and magnetic field) = 90°

3) The magnetic flux (ϕ_m)

when the coil is placed perpendicular to the magnetic field

θ (the angle between the normal to coil's plane and magnetic field) = 0°

When does each of the following values become equal to zero?

3) The magnetic flux (ϕ_m)

when the coil is placed parallel to the magnetic field

θ (the angle between the normal to coil's plane and magnetic field) = 90°

4) The change in magnetic flux ($\Delta\phi_m$)

When the coil rotates one complete cycle from the initial position (beginning of rotation)

5) The average induced emf generated in a coil

When the coil rotates one complete cycle from the initial position (beginning of rotation)

6) Induced emf generated in a conductor (straight wire) moves in a uniform magnetic field

when the conductor moves parallel to the magnetic field lines

θ (the angle between the direction of wire and magnetic field) = zero

7) Induced emf generated in rectangular coil (generator) rotates in a uniform magnetic field

when the coil is perpendicular to the magnetic field

θ (the angle between the normal to coil's plane and magnetic field) = zero

8) The induced emf obtained from the electric transformer

when the primary coil in the transformer connected to DC source

How to increase each of the following:

7) Mutual induction between two coils

a) by inserting of soft iron core inside the coil

b) by increasing number of turns of the secondary coil

c) by decreasing the distance between the two coils

8) Self induction coefficient of a coil

a) by inserting of soft iron core inside the coil

b) by increasing number of turns of the coil

c) by decreasing the distance between the each two turns

9) The induced emf in a straight wire moving in a magnetic field

a) by increasing the magnetic flux density

b) by increasing the length of wire

c) by increasing the velocity of wire

10) the magnitude of the induced current in the coil (induced emf)

- a) Moving the magnet or the coil at a higher speed
- b) Increasing the number of turns on the coil stronger magnet.
- c) Increasing the strength of the magnetic field through the use of a stronger magnet

11) Maximum induced emf in the dynamo

- a) by increasing number of turns of the coil
- b) by increasing the magnetic flux density
- c) by increasing the area of coil
- d) by increasing the frequency

What happens.....? :(Explain each of them briefly)

What are the results produced in each of the following

When the number of turns in the dynamo's coil is doubled and its frequency increases to its double (with respect to the maximum electromotive force generated from it)

Ans: the maximum electromotive force will be 4 times

The reason: according to the relation $\text{emf} = A B N 2\pi f$

the maximum electromotive force is directly proportional to both the number of turns and the frequency

Passing a high alternating current(A.C.) through a coil which surrounds a piece of iron.

Ans: The piece of iron will be heated or may be melt

The reason: due to the generation of eddy current in the surrounded wire because this current has a heat effect

Questions and answers

Mention the three ways to induce an EMF

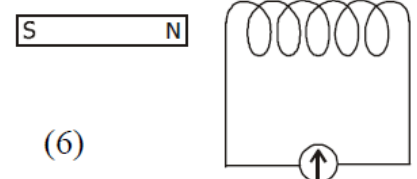
Ans.: a) move the conductor through a magnetic field,
b) move the magnetic field across a conductor,
c) vary the magnetic field strength across a stationary conductor

A coil of wire is connected to a sensitive galvanometer as shown in the diagram.

- a) What is observed when the magnet is moved towards the coil?
- b) Explain why this occurs.
- c) Describe what happens when the speed of the magnet is increased.

Ans

- a) The needle in the galvanometer deflects in certain direction momentarily.
- b) An emf is induced in the coil of wire, which in turn produces a current which moves the needle.
- c) There is a greater deflection of the needle



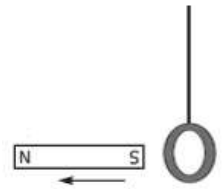
In the shown diagram. A light aluminium ring is suspended from a long thread.

- a) When a strong magnet is moved away from it, the ring follows the magnet. Explain why.
- b) What would happen if the magnet were moved towards the ring?

Ans

a) According to Lenz's law the current flows in the ring in such a direction as to oppose the change which caused it. The face near to the magnet becomes North Pole. Therefore the ring follows the magnet.

b) The ring would be repelled



The factors affecting the magnitude of an induced EMF depends on

Ans: 1) the strength of the magnetic field,
2) the speed of the motion between lines of force and the conductor,
3) the angle between the magnetic lines of force and the conductor
4) the number of turns in the conducting coil

What happens when the speed at which the coil of a generator is rotated is increased?

Ans: Increasing the speed increases the rate at which the flux changes so that increases the emf or the voltage

State Faraday's law of electromagnetic induction.

Ans: The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

If you bring the North Pole of a magnet near a face of a coil, what is the direction of the current induced in that side?

Ans: Anticlockwise.

The reason: This makes that side of the coil magnetically North which repels the magnet coming towards it.

When a glass rod moves perpendicular to a magnetic field, is there any emf induced in it?

Ans: No. Because glass is an insulator

What happens to a velocity of a conductor when it moves in a varying magnetic field?

Ans: Decreases. The eddy currents induced in the conductor damp the motion of the conductor

How can mutual inductance be increased without changing the geometry of the coils?

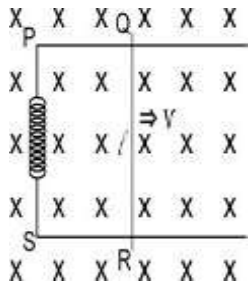
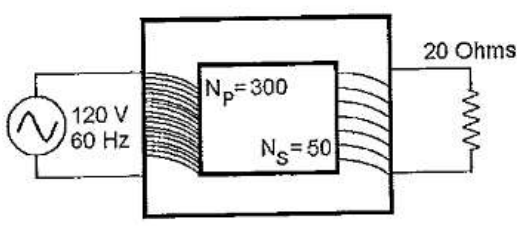
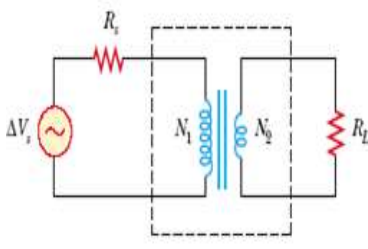
Ans: By inserting a soft iron rod inside the coils

Why does a bulb connected in series with a self-inductance glows brilliantly for a moment when the current in the circuit is switched off?

Ans: Because of the forward emf produced in the coil opposes the change producing it

Problems: (electromagnetic induction)

The magnetic field strength between the poles of a strong electromagnet at a given moment is $B = 0.1 \text{ T}$. A circular coil of radius $r = 2 \text{ cm}$ and having 10 turns is placed within the field and perpendicular to it. If the magnetic field strength increases steadily to 0.25 T in 1.2 ms calculate the voltage induced in the coil while the field is changing	$\text{emf} = -N \frac{\Delta BA}{\Delta t}$ $\text{emf} = -10 \frac{(0.25 - 0.1)(\pi \times (2 \times 10^{-2})^2)}{1.2 \times 10^{-3}}$ $\text{emf} = -1.57 \text{ V}$
A search coil with area 2 cm^2 and 50 turns is attached to a galvanometer and is placed with its plane at right angles to the field between the poles of a magnet. The total circuit resistance is 200Ω. When the coil is suddenly removed from the field the meter measures a total charge flow of $2.4 \mu\text{C}$. Calculate the field strength of the magnet.	$\text{emf} = -N \frac{\Delta BA}{\Delta t}$ $\text{emf} = IR = \frac{Q}{t} R = -N \frac{\Delta BA}{\Delta t}$ if the field vanishes when the coil is suddenly removed from the field $\frac{2.4 \times 10^{-6} \times 200}{t} = \frac{50 \times B \times 2 \times 10^{-4}}{t}$ $B = 0.048 \text{ T}$
Coil of resistance of 10Ω consists 100 turns, each turn has cross section area of 100 cm^2, the plane of the coil is perpendicular (normal) to a uniform magnetic field. If the coil rotates $1/2$ cycle in 0.04 sec and the induced current generated in the coil was 0.5 A, calculate the value of magnetic flux density	$\text{emf} = IR$ $\text{emf} = (0.5)(10) = 5\text{V}$ $\text{emf} = -N \frac{\Delta BA}{\Delta t}$ through $1/2$ cycle $\Delta BA = -BA - BA = -2 BA$ $5 = -100 \frac{-2(B)(100 \times 10^{-4})}{0.04}$ $B = 0.1 \text{ T}$ OR $\text{emf} = I R$ $0.04 = T/2$ $T = 0.08 \text{ sec}$ $f = 1/T = 1/0.08 = 12.5 \text{ Hz}$ $\text{emf}_{\text{ave}} = A B N 4 f$ $5 = 100 \times 10^{-4} \times B \times 100 \times 4 \times 12.5$ $B = 0.1 \text{ T}$
If the emf induced in a solenoid is $1.6 \times 10^{-3} \text{ V}$ Determine self-inductance when the current in it changes from 0 to 1.5 A during the time interval 0.2 s	$\text{emf} = \frac{-L \Delta I}{\Delta t}$ $L = \frac{1.6 \times 10^{-3} \times 0.2}{1.5} = 2.12 \times 10^{-4} \text{ H}$
Two adjacent coils (A) and (B), their number of turns is 200 turns and 10000 turns respectively. If a current of intensity 8 Amperes passes in coil (A), it produces a flux 6×10^{-3} weber in coil (A) and a flux 3×10^{-3} weber in coil (B). Find:	$L = \frac{N \Delta \phi_A}{\Delta I} = \frac{200 \times 6 \times 10^{-3}}{8} = 0.15 \text{ H}$ $M = \frac{N_B \Delta \phi_B}{\Delta I} = \frac{10000 \times 3 \times 10^{-3}}{8}$ $M = 3.75 \text{ H}$

a) The coefficient of self induction of coil A b) The coefficient of mutual induction between them. c) The mean value of the emf in coil (B) when the current vanishes at (A) through 0.02sec.	$\text{emf} = - \frac{N_B \Delta \phi_B}{\Delta t} = \frac{10000 \times 3 \times 10^{-3}}{0.02}$ $\text{emf} = 1500\text{V}$
A 0.5m long metal rod QR completes the circuits as shown in the figure. The area of the circuit is perpendicular to the magnetic field of flux density 0.15T. If the resistance of the total circuit is 3Ω. Calculate the current induced in the rod in the direction as indicated with constant speed of 2ms^{-1}.	$\text{emf} = B l v$ $\text{emf} = (0.15)(0.5)(2)$ $\text{emf} = 0.15\text{ V}$ $\text{emf} = IR$ $0.15 = I \times 3$ $I = 0.05\text{A}$ 
An a.c. generator consist of a coil of 100 turns and cross sectional area of 3m^2, rotating at a constant angular speed at 60 rad/s in a uniform magnetic field of 0.04T ,the resistance of coil is 500 . Calculate a) the maximum current drawn from the generator b) the power dissipated in the coil.	$\text{emf}_{\text{max}} = A B N \omega = 100 \times 0.4 \times 3 \times 60$ $\text{emf}_{\text{max}} = 7200 \text{ volt}$ $\text{emf}_{\text{max}} = I_{\text{max}} R \quad 7200 = I_{\text{max}} \times 500$ $I_{\text{max}} = 14.4\text{A}$ $P = (I)_{\text{eff}} (\text{emf})_{\text{eff}}$ $P = (0.707 \times 14.4)(0.707 \times 7200)$ $P = 51825 \text{ Watt}$
A coil of 300 turns and with an area of 0.05 m^2 is rotated 20 times per second in a field of flux density 0.2 T. Calculate a) the maximum emf produced across the end of the coil	$\text{emf}_{\text{max}} = ABN2\pi f$ $\text{emf}_{\text{max}} = (300)(0.2)(0.05)(2)(\pi)(20)$ $\text{emf}_{\text{max}} = 377\text{V}$
Calculate current drawn by the primary of a transformer which steps down 200V to 20V to operate a device of resistance 20Ω. Assume the efficiency of transformer to be 80%. 	$I_s = \frac{V_s}{R} = \frac{20}{20} = 1\text{A}$ $\frac{\eta}{100} \times \frac{V_p}{V_s} = \frac{I_s}{I_p}$ $\frac{80}{100} \times \frac{200}{20} = \frac{1}{I_p}$ $I_p = 0.125\text{A}$
In the transformer shown in Figure, the load resistor is 50 ohm, The turn's ratio $N_P : N_s$ is 5:2, and the source voltage is 80V. If a voltmeter across the load measures 25 V Calculate efficiency of the transformer? 	$V_s = I_s \times R_s$ $25 = I_s \times 50$ $I_s = 0.5\text{A}$ $\frac{\eta}{100} \times \frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$ $\frac{\eta}{100} \times \frac{80}{25} = \frac{5}{2}$ $\eta = 78.125\%$

A step-up transformer is designed to have an output voltage of 2200 V when the primary is connected across a 110V A.C. Source.

- If the primary has 80 turns, how many turns are required on the secondary
- If a load resistor across the secondary draws a current of 1.5 A, what is the current in the primary, assuming ideal conditions
- If the transformer actually has an efficiency of 95%, what is the current in the primary when the secondary current is 1.2 A?

Ideal transformer

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} \quad \frac{110}{2200} = \frac{80}{N_s}$$

$$N_s = 1600 \text{ turns}$$

$$\frac{I_p}{I_s} = \frac{N_s}{N_p} \quad \frac{I_p}{1.5} = \frac{1600}{80}$$

$$I_p = 30 \text{ A}$$

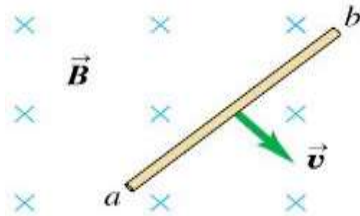
If the efficiency = 95%

$$\frac{\eta}{100} \times \frac{V_p}{V_s} = \frac{I_s}{I_p}$$

$$\frac{95}{100} \times \frac{110}{2200} = \frac{1.2}{I_p}$$

$$I_p = 25.26 \text{ A}$$

When a thin 18cm iron rod moves with a constant velocity of 3.5 m/s perpendicular to the rod in the direction shown in the figure, the induced emf across its ends is measured to be 0.45V.



- What is the magnitude of the magnetic field?
- Which point is at a higher potential, a or b?

$$\text{emf} = B l v$$

$$0.45 = (B) (0.18) (3.5)$$

$$B = 0.714 \text{ T}$$

According to Fleming's right hand rule

The induced current will be from a to b
Point **b** has higher potential difference than point **a**

The mutual inductance between a primary and secondary circuits is 0.5 H. The resistances of the primary and the secondary circuits are 20 ohms and 5 ohms respectively. To generate a current of 0.4 A in the secondary, calculate the rate of current in the primary

$$M = \frac{\text{emf}_s}{\frac{\Delta I_p}{\Delta t}} = \frac{I_s \times R_s}{\frac{\Delta I_p}{\Delta t}}$$

$$0.5 = \frac{0.4 \times 5}{\frac{\Delta I_p}{\Delta t}}$$

$$\frac{\Delta I_p}{\Delta t} = \frac{2}{0.5} = 4 \text{ A/s}$$